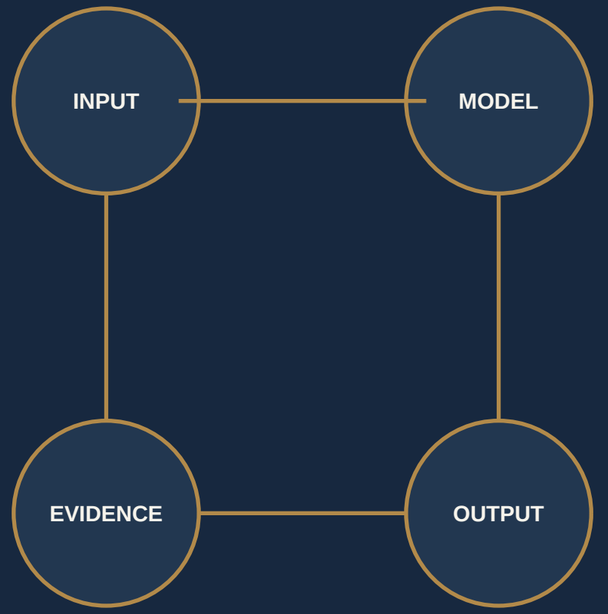
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Design

AI.411 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define the complete robot-design process: requirements engineering.
- Apply key methods including mechanical structure.
- Build or analyze a practical system involving actuators.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

The complete robot-design process: requirements engineering

02

METHODS

Mechanical structure

03

SYSTEMS

Actuators

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · The complete robot-design pro

The complete robot-design process: requirements engineering

02 · Mechanical structure

Mechanical structure

03 · Actuators

Actuators

04 · Sensors

Sensors

05 · Electronics

Electronics

06 · Control architecture

Control architecture

07 · Software

Software

08 · Power systems

Power systems

Core ideas, made explicit

AI.411 • Distinguish definitions, assumptions and evidence

- The complete robot-design process: requirements engineering
- Mechanical structure
- Actuators
- Sensors
- Electronics

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.411 • Distinguish definitions, assumptions and evidence

- Control architecture
- Software
- Power systems
- Safety
- Manufacturability and testing

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM

The complete robot-design process: requirements engineering

INPUTS

Mechanical structure

METHOD

Actuators

OUTPUT

Sensors

EVALUATION

Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving the complete robot-design process: requirements engineering.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using the complete robot-design process: requirements engineering.
- Implement or simulate a baseline using mechanical structure.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.